

MINISTRY OF EDUCATION AND TRAINING
EASTERN INTERNATIONAL UNIVERSITY



MIS 395
ARTIFICIAL INTELLIGENCE FOR BUSINESS

Final Project

**Multidomain Analytics for Strategic
Decision-Making**

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I. Project Motivation

As artificial intelligence (AI) continues to transform the business landscape, its integration into strategic decision-making processes is becoming increasingly essential. This project is motivated by the need to harness AI-powered tools across diverse business domains - namely marketing, retail operations, and human resources - to uncover actionable insights and drive performance improvements.

In the marketing domain, we utilize the Bank Marketing dataset from the UCI Machine Learning Repository, which provides detailed records of direct marketing campaigns. The application of AI to this data enables organizations to predict customer responses, enhance targeting strategies, and improve campaign effectiveness.

In the realm of retail analytics, the Bike Sales dataset from Kaggle offers over 100,000 transaction records. AI techniques applied to this data allow businesses to detect sales patterns, forecast demand, and optimize inventory planning - critical factors in maintaining operational efficiency and profitability.

In human resource management, the Employee Performance and Productivity dataset provides insights into workforce behavior and outcomes. Through AI-driven analysis, we aim to identify performance trends, evaluate productivity drivers, and support evidence-based talent management strategies.

By combining these three datasets, this project highlights the cross-functional value of AI and its potential to streamline workflows, personalize services, and reduce operational costs. Each dataset represents a unique business scenario where AI can assist in making faster, data-informed decisions. The project ultimately seeks to demonstrate that AI is not just a technological trend but a fundamental enabler of modern business intelligence.

II. Dataset Overview

No	Dataset Name	Domain	No. of Records	No. of Features	Model Applied	Missing Value?
1	Bank Marketing Campaign Data	Marketing	45,211	16	Classification	No
2	Bike Sales Data	Retail	100,000	11	Regression	No

3	Employee Performance & Productivity	Human Resources	100,000	20	Classification	No
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III. Tools Used and Why

No	Tools	Reasons
1	Python	Offers full control for end-to-end analysis, from data preprocessing to modeling and evaluation. Supports a wide range of libraries (e.g., pandas, scikit-learn, matplotlib) that enable powerful EDA and predictive modeling capabilities.
2	Graphite Note	Provides a structured, user-friendly interface for building models and generating automated business insight reports. Enables non-code users to explore AI modeling while maintaining high-quality outputs suitable for stakeholder interpretation.
3	PartyRock	Allows for fast prototyping and scenario testing through prompt-based interaction without the need for traditional model training. While results may vary based on prompt quality, it is effective for generating initial ideas and exploring potential business outcomes quickly.

IV. Key Findings (with Visuals) for Bank Marketing Campaign Data

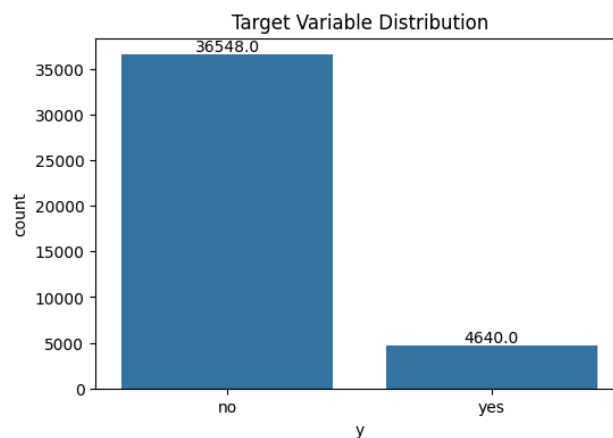
1. Descriptive by Python:

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
age	41188.0	NaN	NaN	NaN	40.02406	10.42125	17.0	32.0	38.0	47.0	98.0
job	41188	12	admin.	10422	NaN	NaN	NaN	NaN	NaN	NaN	NaN
marital	41188	4	married	24928	NaN	NaN	NaN	NaN	NaN	NaN	NaN
education	41188	8	university.degree	12168	NaN	NaN	NaN	NaN	NaN	NaN	NaN
default	41188	3	no	32588	NaN	NaN	NaN	NaN	NaN	NaN	NaN
housing	41188	3	yes	21576	NaN	NaN	NaN	NaN	NaN	NaN	NaN
loan	41188	3	no	33950	NaN	NaN	NaN	NaN	NaN	NaN	NaN
contact	41188	2	cellular	26144	NaN	NaN	NaN	NaN	NaN	NaN	NaN
month	41188	10	may	13769	NaN	NaN	NaN	NaN	NaN	NaN	NaN
day_of_week	41188	5	thu	8623	NaN	NaN	NaN	NaN	NaN	NaN	NaN
duration	41188.0	NaN	NaN	NaN	258.28501	259.279249	0.0	102.0	180.0	319.0	4918.0
campaign	41188.0	NaN	NaN	NaN	2.567593	2.770014	1.0	1.0	2.0	3.0	56.0
pdays	41188.0	NaN	NaN	NaN	962.475454	186.910907	0.0	999.0	999.0	999.0	999.0
previous	41188.0	NaN	NaN	NaN	0.172963	0.494901	0.0	0.0	0.0	0.0	7.0
poutcome	41188	3	nonexistent	35563	NaN	NaN	NaN	NaN	NaN	NaN	NaN
emp.var.rate	41188.0	NaN	NaN	NaN	0.081886	1.57096	-3.4	-1.8	1.1	1.4	1.4
cons.price.idx	41188.0	NaN	NaN	NaN	93.575664	0.57884	92.201	93.075	93.749	93.994	94.767
cons.conf.idx	41188.0	NaN	NaN	NaN	-40.5026	4.628198	-50.8	-42.7	-41.8	-36.4	-26.9
euribor3m	41188.0	NaN	NaN	NaN	3.621291	1.734447	0.634	1.344	4.857	4.961	5.045
nr.employed	41188.0	NaN	NaN	NaN	5167.035911	72.251528	4963.6	5099.1	5191.0	5228.1	5228.1
y	41188	2	no	36548	NaN	NaN	NaN	NaN	NaN	NaN	NaN

The dataset contains 41,188 records with 17 variables covering client demographics, financial status, and marketing campaign details. The average client age is 40, most are married and work in administrative roles, and over half have housing loans.

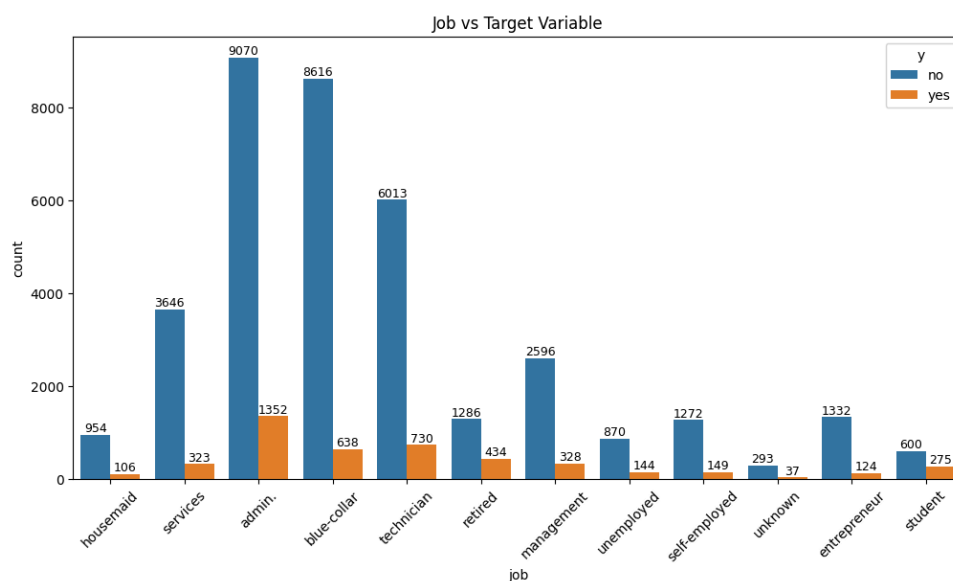
2. EDA by Python:

a. Target Variable Distribution



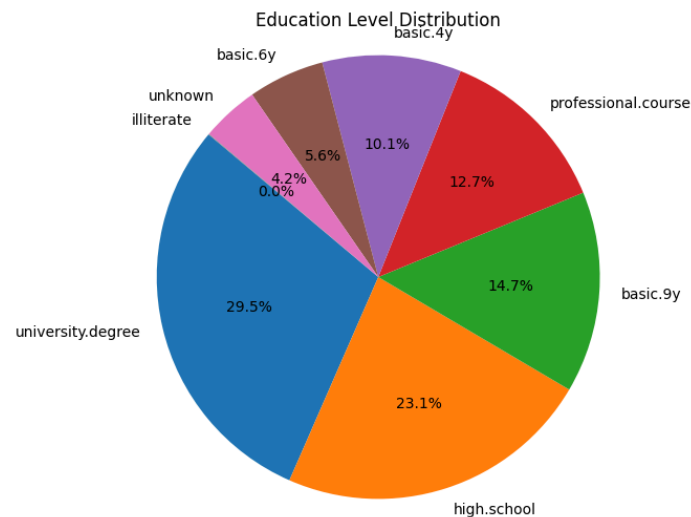
The target variable y exhibits a pronounced class imbalance. Approximately 88.7% of instances correspond to the “no” category, indicating that the majority of clients did not subscribe to the offered product, whereas only 11.3% fall into the “yes” category. Such imbalance has significant implications for predictive modeling, as standard classification algorithms may become biased towards the majority class.

b. Job vs Target Variable



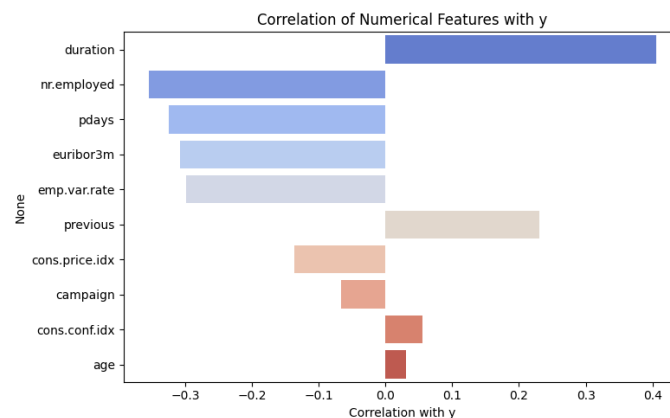
The distribution of subscription outcomes across different occupational groups reveals both volume and propensity patterns. Occupations such as “admin.,” “blue-collar,” and “technician” account for the largest shares of the dataset but exhibit relatively low subscription rates. Conversely, groups such as “student,” “retired,” and “unemployed” display notably higher subscription rates despite their smaller representation.

c. Education Level Distribution



The educational attainment of clients is skewed towards higher levels, with “university degree” (29.5%) and “high school” (23.1%) being the most prevalent categories. Intermediate education levels (e.g., “basic.4y,” “basic.6y,” “basic.9y”) constitute smaller proportions, while “illiterate” is virtually absent.

d. Correlation of Numerical Features with y



Among numerical variables, “duration” demonstrates the strongest positive correlation with the target variable (approximately 0.4), indicating that longer call durations are associated with a higher likelihood of subscription. However, as duration is determined during the call, it cannot serve as a predictive feature in pre-contact modeling. Other variables, including “nr.employed”, “euribor3m” and “emp.var.rate”, exhibit weak negative correlations with y , suggesting that macroeconomic and labor-related factors may exert a modest inverse influence on subscription likelihood.

e. Subscription Rate by Contact Type

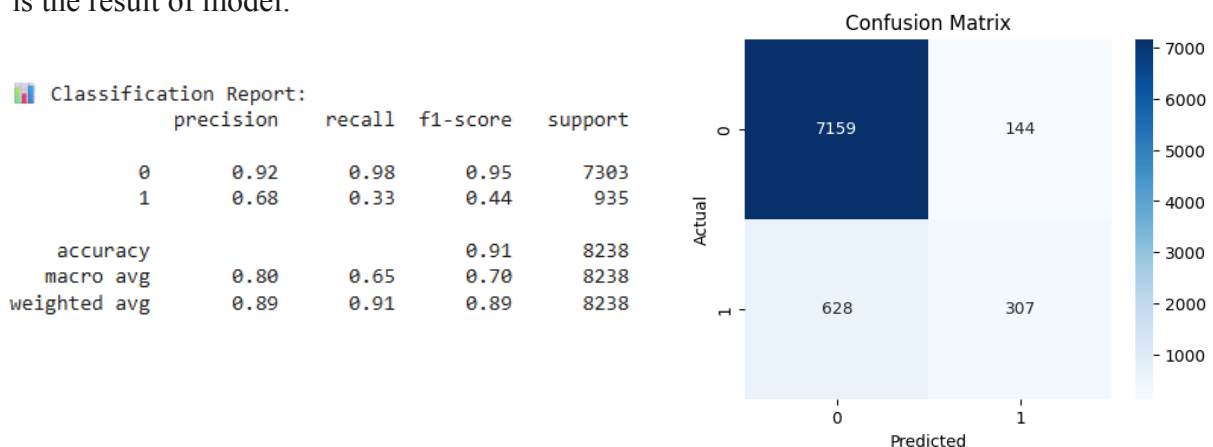
Yes Rate (%)	
contact	
cellular	14.74%
telephone	5.23%

A pronounced difference emerges between the two available contact methods. Clients reached via cellular devices display a subscription rate of 14.74%, nearly three times higher than those contacted via telephone landlines (5.23%). This disparity may reflect generational differences, accessibility, or the immediacy of mobile communication.

3. Models by Python and Graphite Note:

a. Python (Logistic Regression and Random Forest)

Logistic Regression was selected for this dataset because the target variable is binary, making it a natural fit for a classification algorithm that estimates the probability of class membership. The model is interpretable, allowing coefficients to be directly related to feature importance, which is valuable for understanding the drivers of customer subscription. Below is the result of model:



The classification report and confusion matrix indicate that the model achieves an overall accuracy of 91%. However, this high accuracy is primarily driven by strong performance in predicting the majority class (label 0), which corresponds to clients who did not subscribe to the product. For class 0, the model attains a precision of 0.92 and recall of 0.98, resulting in an F1-score of 0.95.

In contrast, the minority class (label 1), representing clients who subscribed, shows notably weaker results. The precision is relatively high at 0.68, indicating that when the model predicts a subscription, it is correct in most cases. However, recall is low at 0.33, meaning the model fails to identify a substantial portion of actual subscribers. The corresponding F1-score is 0.44, reflecting this imbalance between precision and recall.

The confusion matrix confirms these findings: out of 935 actual subscribers, only 307 are correctly identified (true positives), while 628 are missed (false negatives). This suggests that the model is biased towards predicting the majority class, a common issue with imbalanced datasets such as this one.

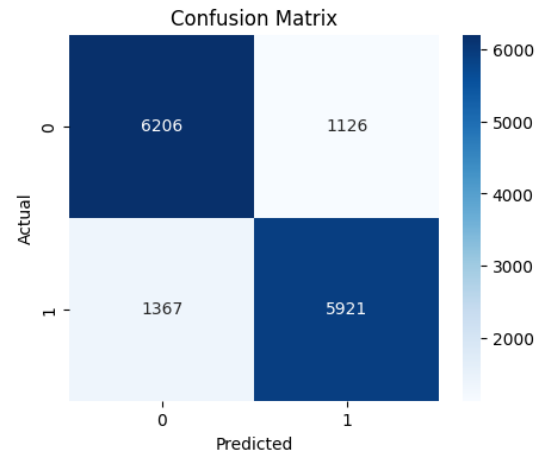
Overall, while the model is effective at identifying non-subscribers, its limited ability to capture true subscribers may hinder the campaign's targeting effectiveness. Strategies such as class weighting, oversampling (SMOTE), or alternative algorithms may improve minority class recall and lead to a more balanced predictive performance.

Following the initial logistic regression modeling, the results revealed a strong bias toward the majority class, with limited recall for actual subscribers. To address this imbalance and enhance predictive performance, the subsequent workflow expands into a full classification pipeline incorporating exploratory data analysis (EDA), extended feature engineering, and advanced model tuning. This enhanced approach leverages SMOTE to balance class distributions, evaluates both Logistic Regression and Random Forest classifiers, and employs ROC curve analysis to compare model discriminative power. The objective is to mitigate the shortcomings of the initial model, improve recall for the minority class, and identify a more robust solution for deployment.

✓ Accuracy: 0.8294801641586868

Classification Report:

	precision	recall	f1-score	support
0	0.82	0.85	0.83	7332
1	0.84	0.81	0.83	7288
accuracy			0.83	14620
macro avg	0.83	0.83	0.83	14620
weighted avg	0.83	0.83	0.83	14620

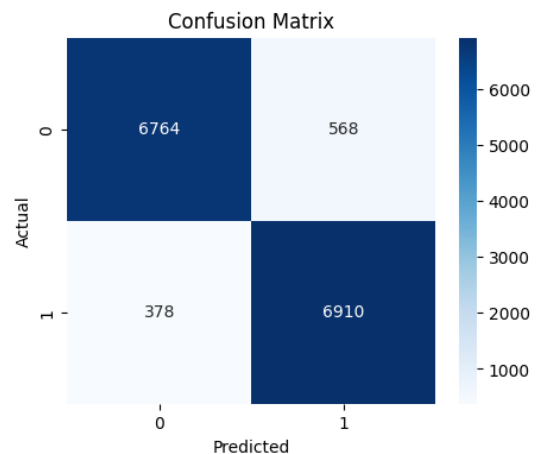


With SMOTE balancing, Logistic Regression achieves 82.95% accuracy and nearly equal performance for both classes (F1-score = 0.83 for each). The confusion matrix shows 6,206 true negatives, 5,921 true positives, and a symmetrical distribution of errors (1,126 false positives, 1,367 false negatives). This balance marks a clear improvement in minority-class recall, making the model more suitable for deployment when both classes are equally important.

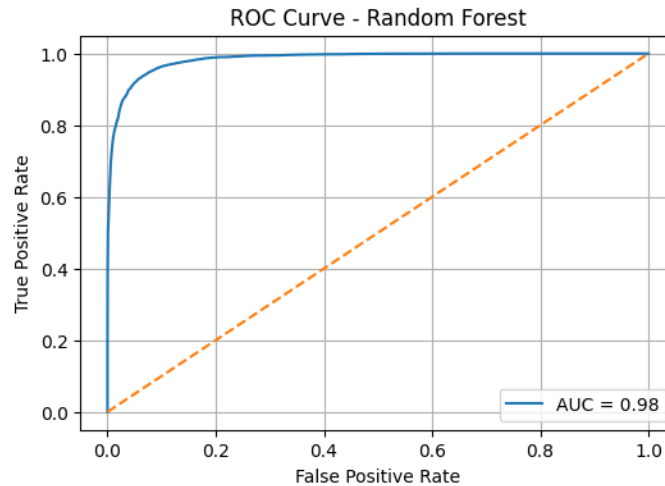
✓ Accuracy: 0.9352941176470588

Classification Report:

	precision	recall	f1-score	support
0	0.95	0.92	0.93	7332
1	0.92	0.95	0.94	7288
accuracy			0.94	14620
macro avg	0.94	0.94	0.94	14620
weighted avg	0.94	0.94	0.94	14620



With SMOTE balancing, Random Forest reaches 93.53% accuracy, combining high precision and recall for both classes (Class 0: F1 = 0.93; Class 1: F1 = 0.94). The confusion matrix shows 6,764 true negatives and 6,910 true positives, with minimal errors (568 false positives, 378 false negatives). This balance, coupled with low misclassification rates, demonstrates superior performance over Logistic Regression, making Random Forest the preferred choice for deployment on this dataset.



The ROC curve for the Random Forest model shows an exceptional area under the curve (AUC = 0.98), indicating excellent discriminative ability between the subscriber and non-subscriber classes. The curve rises steeply towards the top-left corner, which reflects a very high true positive rate coupled with a very low false positive rate across a wide range of classification thresholds.

This performance confirms that the model maintains strong separation between the two classes, even when threshold adjustments are applied. The high AUC score suggests that the Random Forest classifier is not only accurate at the chosen threshold (as evidenced by the confusion matrix and classification report) but also robust and adaptable to different operational requirements where the balance between sensitivity and specificity might need to be tuned.

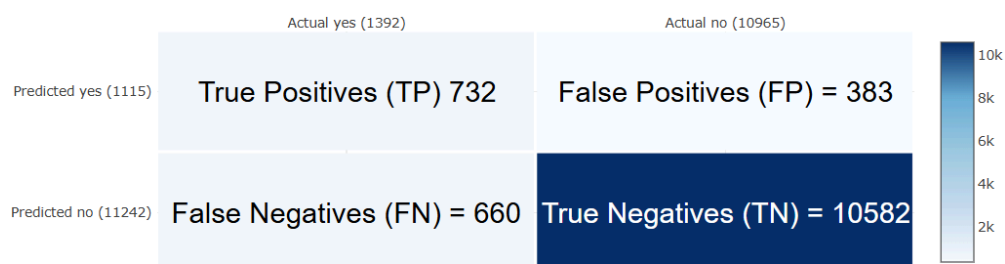
Combined with its high precision, recall, and F1-scores for both classes, the ROC curve reinforces the conclusion that Random Forest - with SMOTE preprocessing - offers a superior and highly reliable predictive framework for this dataset.

b. Graphite Note (Light GBM)

Model	F1	Accuracy	AUC	MCC	Precision	Recall	
Light GBM	91.23%	91.56%	94.71%	54.79	91.02%	91.56%	🌟
Gradient Boosting	90.99%	91.45%	94.45%	53.24	90.75%	91.45%	
Random Forest	90.57%	91.23%	93.78%	50.69	90.32%	91.23%	
Logistic Regression	90.17%	91.05%	93.43%	48.32	89.96%	91.05%	

The results show that Light Gradient Boosting Machine (Light GBM) delivers the best performance, with the highest F1-score (91.23%), accuracy (91.56%), and AUC (94.71%). Its balanced precision (91.02%) and recall (91.56%) indicate strong capability in correctly identifying both classes.

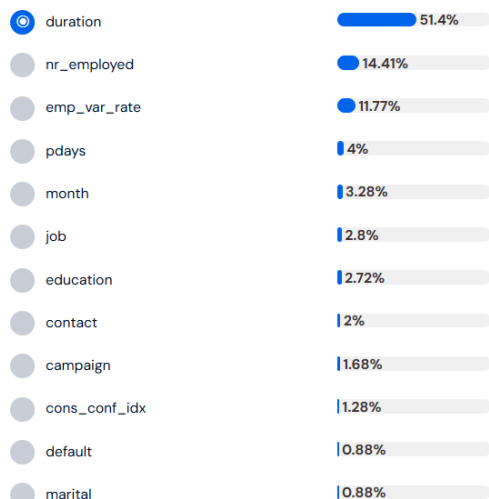
Gradient Boosting and Random Forest follow closely, each exceeding 90% in F1-score, accuracy, and AUC, while Logistic Regression also performs well despite being a simpler model. In contrast, models like Decision Tree and K Neighbors show lower scores, suggesting weaker generalization.



The model correctly predicted 732 true positives and 10,582 true negatives, showing strong overall accuracy. However, 660 false negatives indicate missed actual subscribers, while 383 false positives could lead to unnecessary follow-ups. Improving recall for the positive class could help capture more true subscribers.

Key Drivers refer to how much this Model relies upon each column (feature) to make accurate predictions. The more a model relies on a column (feature) to make predictions, the more important it is for the Model. Graphite uses a permutation feature importance for this calculation.

The most important key driver to predict **y** is **duration**. The least important is **housing**.



Explore insights on how **y** outcomes are affected by the changes in **duration** values.

When duration is..	Likelihood of y being yes
⑦ (510.0, 551.0]	Increases by →3.23x
⑦ (469.0, 510.0]	Increases by →2.88x
⑦ (428.0, 469.0]	Increases by →2.45x
⑦ (387.0, 428.0]	Increases by →2.25x
⑦ (346.0, 387.0]	Increases by →1.8x
⑦ (305.0, 346.0]	Increases by →1.69x

The analysis shows that duration is by far the most influential feature in predicting y, contributing 51.4% to the model's decisions. This is followed by "nr_employed" (14.41%) and emp_var_rate (11.77%), while housing has the least impact (0.88%).

For "duration", longer call times significantly increase the likelihood of a positive outcome. Calls lasting 510–551 seconds raise the chance of a "yes" by 3.23x, while those in the 469–510 second range increase it by 2.88x. Even moderate durations (305 to 346 seconds) still improve the likelihood by 1.69x, reinforcing that engagement time is a strong driver of conversions.

These results suggest that maintaining longer, meaningful customer interactions is a key success factor, and that workforce and economic indicators ("nr_employed", "emp_var_rate") also play a role, albeit to a lesser extent.

Future improvements could involve adding demographic, behavioral, temporal, and external economic features to enhance predictive accuracy. Overall, the model is stable, generalizes well, and provides actionable insights that can be integrated into business operations via Graphite Note for improved decision-making.

Actionable Insights from Graphite Note:

a. Key Insights from Model Analysis

Top Predictors:

- ◆ duration (51.4% importance): Calls over 428 seconds, especially 469–551s, greatly increase conversion likelihood.
- ◆ nr_employed (14.41%) & emp_var_rate (11.77%): Moderate employment regions and low/negative variation rates correlate with higher responses.
- ◆ pdays: New leads (no recent contact) respond better.
- ◆ Best Months: March, September, October.
- ◆ Demographics: Students, retirees, and university-educated customers respond more often.
- ◆ Contact Method: Cellular > telephone.
- ◆ Other: Fewer call attempts improve success; low consumer confidence can boost receptiveness.

b. Strategic Pillars

Target High-Value Segments: Focus on students, retirees, university-educated, and fresh leads from favorable employment regions.

Optimize Engagement Quality: Train agents to sustain calls over 428s; shift contact to mobile channels.

Time Campaigns Strategically: Concentrate efforts in March, September, and October.

Manage Contact Frequency: Cap follow-ups at six attempts to prevent fatigue.

Leverage Economic Context: Adjust targeting and messaging during low consumer confidence or negative employment variation periods.

c. Objectives & KPIs

Raise conversion rate by 15–20% next cycle.

Track: positive response rates by segment, % calls in optimal duration range, conversion by contact method, performance by month, and correlation with economic indicators.

d. Recommended Actions

- ❖ Prioritize fresh, high-probability leads.
- ❖ Extend meaningful call duration through training and KPIs tied to conversation quality.
- ❖ Maintain and validate mobile contact databases.
- ❖ Schedule campaigns in high-performing months.
- ❖ Integrate economic data into lead scoring for dynamic targeting.
- ❖ Personalize scripts based on education and occupation.

Final Note: Executing these recommendations requires cross-functional coordination, continuous KPI monitoring, and iterative refinement. By focusing on engagement quality, demographic targeting, and contextual timing, campaigns can achieve higher ROI and stronger customer relationships.

4. AI by PartyRock:

Bank Deposit Subscription Predictor – Final Project

Bank Marketing Predictor Introduction

This tool predicts whether a customer will subscribe to a term deposit based on their profile and campaign interaction history. Fill in the information below to get a detailed prediction and analysis.

Personal Information Section

Personal and Professional Details

Customer Age

→ Enter customer age (e.g., 35)

Job Type

Select job type

Marital Status

Select marital status

Education Level

Select education level

Credit Default

no

Financial Information Section

Financial Status and Loans

Campaign Information Section

Contact and Campaign Details

Contact Method

Select contact method

Contact Duration

→ Enter last contact duration in seconds

Campaign Contacts

→ Number of contacts performed

Days Since Contact

→ Days since last contact (-1 if never contacted)

Euribor Rate

→ Enter Euro Interbank Offered Rate (3-month)

Previous Outcome

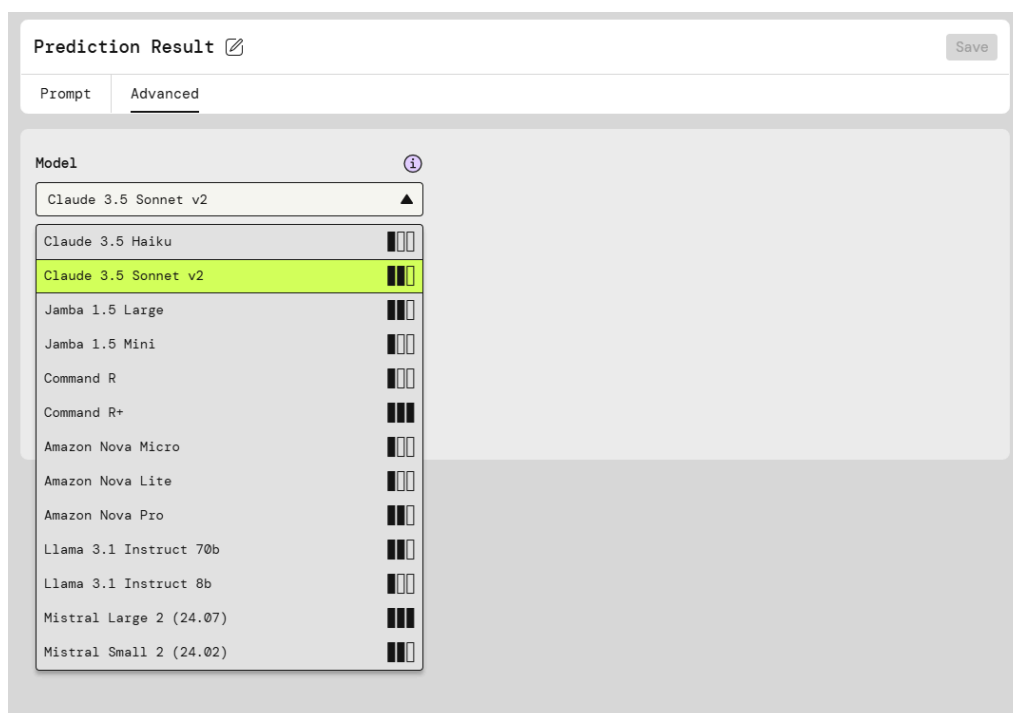
Outcome of previous campaign

Prediction Result

Fill in **Customer Age**, **Job Type**, **Marital Status**, **Education Level**, **Account Balance**, **Loan Status**, **Credit Default**, **Contact Method**, **Campaign Contacts**, **Days Since Contact**, **Euribor Rate** and **Previous Outcome** to get started!

The prompt for PartyRock is designed to clearly instruct the AI on building an app that predicts whether a customer will subscribe to a term deposit. It starts by specifying the exact purpose of the app—binary prediction of “yes” or “no” outcomes in a banking marketing context—ensuring the AI understands the business goal. The prompt then provides a comprehensive list of input fields, including customer demographics (e.g., age, job, marital status, education), financial details (e.g., account balance, loan status, credit default), and campaign interaction metrics (e.g., contact method, last call duration, previous campaign outcome). For each input, the expected format (number input, dropdown) is explicitly stated, which helps PartyRock create a well-structured and user-friendly interface.

Additionally, the prompt instructs the AI to present the prediction in a simple, business-oriented format, avoiding raw probability scores and instead returning a clear “Yes” or “No.” It also requests that the app include a brief explanation of why a particular outcome was predicted, turning the tool into both a prediction engine and a decision-support system. By excluding CSV uploads and focusing on direct value inputs, the prompt ensures the app is lightweight and accessible, suitable for quick, on-demand predictions without data preprocessing requirements. This structure makes the prompt effective in producing an app that is both functional for analysts and easy to use for banking professionals.



A major limitation of PartyRock is that it does not actually train a model on the user’s dataset. Instead, it relies on pre-existing AI models (such as Claude, Command R, Amazon Nova, Llama, Mistral, etc.) to generate predictions based on the prompt. This means the output is not derived from learning directly on the project’s real data but rather from the model’s general knowledge and inference capabilities. As a result, predictions can be prone to inaccuracies, with the AI potentially making assumptions or applying irrelevant information that does not align with the specific dataset, reducing objectivity and reliability.

For this reason, PartyRock is not an ideal tool for projects such as the bank deposit subscription prediction task. Without direct training and evaluation on the processed, domain-specific dataset, the accuracy and trustworthiness of results are significantly

diminished. Data-driven projects requiring precise and unbiased predictions should instead use platforms that allow model training, fine-tuning, and validation on the actual dataset to ensure results truly reflect the data's characteristics and minimize subjective outputs.

V. Tool Comparison

When comparing the tools in terms of usability, speed, visualization capacity, and flexibility, ***Python provides the highest level of flexibility and control*** due to its coding-based nature. Users can customize every step of the pipeline, from preprocessing to model tuning, and integrate multiple libraries for specific needs. However, this flexibility comes at the cost of a steeper learning curve and longer development time compared to no-code solutions. Graphite Note, as a no-code tool, excels in speed and visualization capacity, offering ready-to-use charts, feature importance analyses, and model comparisons without requiring programming knowledge. PartyRock, while very quick to set up and use, is limited in flexibility because it relies on pre-built AI models and cannot directly train on the provided dataset, making it less suitable for precise, data-driven projects.

From a user experience perspective, Graphite Note is intuitive for non-technical users and provides an accessible entry point for machine learning projects. Its guided workflows and automated reports simplify complex tasks, enabling quick iteration. Python, on the other hand, offers a richer and more customizable experience for advanced users who are comfortable coding, but beginners may find it challenging without prior experience. PartyRock delivers the easiest learning curve overall since it focuses on generating outputs directly from prompts, but this simplicity sacrifices transparency, control, and accuracy.

When evaluating which tool suits which task, Python is clearly better suited for coding-based projects that require complete control, rigorous validation, and reproducibility. Graphite Note fits scenarios where the goal is to quickly build, test, and visualize models with minimal coding, making it ideal for business analysts or teams without strong programming skills. ***PartyRock, due to its reliance on pre-trained language models rather than dataset-specific training, is not recommended*** for critical prediction tasks like the bank deposit subscription project, but could be useful for prototyping ideas, generating narratives, or creating concept demonstrations where high accuracy is not essential.

VI. Business Insights & Conclusion

These tools provide significant value to businesses by enabling data-driven decision-making in areas such as marketing, inventory management, and staffing.

For example, Python and Graphite Note can process and analyze large volumes of customer and campaign data to identify high-potential leads, allowing companies to focus marketing resources where they are most likely to yield conversions. This targeted approach improves campaign efficiency, reduces wasted effort, and ultimately boosts ROI. Similarly, predictive models built with these tools can forecast demand trends, helping optimize inventory levels and reduce stockouts or overstock situations.

In staffing decisions, these tools can analyze historical performance and operational data to predict peak demand periods, enabling more accurate workforce planning.

While Python offers maximum flexibility for creating highly customized solutions, Graphite Note provides quick, accessible insights through its no-code interface, making advanced analytics available to non-technical teams. By integrating such predictive analytics into everyday operations, businesses can respond proactively to market changes, improve efficiency, and maintain a competitive edge.